

Final Report

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Part I

We first learned the concepts of artificial intelligence and briefly introduced the history of artificial intelligence. Then we tried the first AI model, which is used for image classification. We first train the model with our own input data with labels, and then the model can identify the given image whose label is based on the training data. If our input data is enough, the model can make sure to identify all the images correctly. Then we briefly introduced the difference between artificial intelligence and machine learning.

A very important concept in machine learning is performance and performance measures, which evaluates how well a model performs on a task. A performance measure can be different based on which kind of task we are going to do. For example, on a recognition learning problem, the performance measure can be the classification accuracy, which is the number of correct outputs divided by the total number of examples. On the function fitting problem, the performance measure can be the mean squared error, which represents the total distance from the points on the fitted curve to the input data points.

Another very important concept in machine learning is experience, which is what kind of input data you are going to feed the model for learning to improve the performance for the corresponding task. For example, on a recognition learning problem, the experience is the data that you want to recognize with its corresponding labels. On the function fitting problem, the experience is the data points along with their function values. Usually, the samples or input data will be split into three datasets, which are the training set, validation set, and test set. The samples in the training set are fed to the program for the learning task. After the training process, the validation set is used to fine-tune the program. Finally, the test set is used to evaluate if the program is learning well.

Another very important concept is underfitting and overfitting. Underfitting means the input data is too few, so that the model does not learn much from the input data and then leads to a bad performance. Overfitting means that the input data is too much, so the model learned too much from the input data, and then the model can only work with the input data without any change, which means if the given data is not the same as the input data or has a few changes, the model might not work.

There are three different machine learning tasks, which are regression, classification, and clustering. The regression task is to predict continuous values. The classification task is to

predict discrete values. The clustering task is to determine the structure of data. There are four main machine learning types, which are supervised learning, unsupervised learning, semi-supervised learning, and reinforcement learning. Supervised learning means all the input data has its corresponding label. Regression tasks and classification tasks are supervised learning problems. Unsupervised learning means that instead of labeled input data, it looks for interesting patterns in the data. Clustering tasks and dimensionality reduction are unsupervised learning tasks. Semi-supervised learning means that only some of the input data has its corresponding label. Reinforcement learning means that the model interacts with the world and learns to maximize a scalar reward signal.

Then we learned some basic concepts of machine learning, such as linear classifiers, geometric margin, support vector machines, gradient descent, stochastic gradient descent, and linear regression. We also learned what is convolutional neural network (CNNs) and transformer language models. Basically, the transformer language models have an encoder and a decoder. The encoder processes the input data based on the MultiHead layer and the feedforward attention layer. The decoder generates the output data word by word based on the encoder and its previous outputs using the same MultiHead layer and the feedforward attention layer.

We also learned what the Internet is. The internet is built by network devices, network edge, and network core. There is a packet switch that can break the message into a packet and then send it to other network devices. In the wireless network, there are six main areas that we need to consider, which are mobility, user data rate, area traffic capacity, energy efficiency, latency, and device density. There are four main elements in a wireless network, which are the wireless hosts, base station, wireless link, and wireless device radio. For the radio link, there are four main categories, which are wireless local area network, wide area network, Bluetooth, and satellite. A 5G radio access network is an edge network connecting devices to a base station. Software-defined radio access network is an implementation of a radio access network using a software-defined networking approach. A challenge in networking is mobility. When an iPhone needs to move from home to school, its iPhone will move from the current base station at home to the target base station at school. During this process, the current base station and target base station need to do a lot of work so that the iPhone cannot immediately get on the network. This is where the AI can help. AI can help learn the patterns of users' behavior to predict the future movement so that the base station can do the handover earlier.

Probability is a very important part of AI, because it gives AI a tool to make optimal decisions. Probability mass functions are a probability distribution over discrete variables, such as fair coin flips and the Bernoulli distribution. Probability density functions are a

probability distribution over continuous variables, such as bus arrival and the Gaussian distribution. Maximum likelihood estimation is a tool to estimate unknown parameters using observed data. We also learned about the Gaussian distribution, conditional probability, Bayes' rule, and entropy and information theory.

Generative models are a very important class of models in machine learning. Statistical generative models are learned from data. Basically, the statistical generative model is a probability distribution, and it generates data based on the probability distribution. There are three main categories of machine learning problems, which are discriminative, reinforcement, and generative, with an overlap between all of them. Discriminative means the model needs to classify the data. Generative means the model needs to generate data. This means that the discriminative model needs a label for the data to do the classification task, but the generative model does not need a label for the data. Variational Auto-Encoder is an example of a generative model and simply uses a feed-forward neural network, which means it has fast sampling and model coverage or diversity. The generative adversarial network is another example of a generative model, and it is the only likelihood-free generative model, which means it has fast sampling and high-quality samples. The denoising diffusion model is another example of a generative model, and it has high-quality samples and model coverage or diversity.

A language model is a probabilistic characterization of the tokens. Each token comes from a finite number of categories, and the categories are not ordered. A generative model for text is a language model that can generate text. An autoregressive model is a generative model where the next token only depends on previous tokens, which performs well when the data is sequential, such as speech or text. A transformer is a model that can find the global dependencies between input and output, and it can generate a contextual word embedding. The transformer language model is also called large language model, which can be split to three different main categories: BERT-like (encoder-only) models, GPT-like (decoder-only) models, and combined models.

Right now, for most of the models, the datasets are huge, and computational resources require a lot, so training a model needs a lot of time. This is where the distributed learning comes in. The distributed learning splits both data and model to compute them across nodes. The distributed learning can be split into two paradigms, which are federated learning and decentralized learning. The federated learning has a server, and the server will send data to other clients. The clients will perform local computation based on the data that the server sent and then send the performance back to the server. After the server gets all the results from the clients, it will aggregate the results and update the model. However, there is a problem that if the server has a failure or crash, all system will fail. Decentralized

learning doesn't have a server, so if one node fails, it will not affect all systems, and since there is no server, it is very hard to be attacked.

Part II

The paper I selected is "[What Is ChatGPT Doing and Why Does It Work?](#)". Basically this paper is talking about how ChatGPT generates a correct answer. Instead of generating a correct answer, it is based on the current text to predict what the next word will be, which means it will just be based on the probability of the next word. To be more accurate, it will add one token, but one token means 4 characters in English, which might be a part of a word, and this is why sometimes ChatGPT will generate a new word. However, this does not mean ChatGPT will always choose the word that has the highest probability. Because if it always chooses the word that has the highest probability, it will always generate the same answer if the question does not change, which eliminates the creation of ChatGPT. Therefore, ChatGPT will add some randomness to make it choose other words so that it will generate a different answer for the same question, and this randomness is called temperature.

The optimal value for temperature is 0.8 based on the fact, but there is no theory to prove why 0.8 is the best. The probability of words comes from an estimate of how common each word is in a large amount of English text, which can be a few million books, with altogether a few hundred billion words. But since all the words are randomly selected and the probability of a word appearing is the same as the probability in the English text that we trained. To address this issue, we will not only consider the probability of each word but also the probabilities of pairs or longer n -grams of words. Since there are only 40,000 commonly used words, the number of possible 2-grams is already 1.6 billion, and even the number of possible 3-grams is 60 trillion. It is impossible to estimate the probabilities of all the words based on the current text, so we need to build a model that can estimate the probabilities with which sequences should occur. ChatGPT is based on a large language model that can estimate those probabilities very well.

There is never a model-less model, which means all models have used some structure and a set to tune the model, such as parameters. For example, ChatGPT used 175 billion of them. ChatGPT is basically just a huge neural network. The neural network is a combination of many neurons, but each neuron has a simple function. Based on traditional biological neurons, each neuron has a certain connection between the current neuron and the previous neuron, which will be assigned a weight. The input value current neuron will be the weight multiplied by the value from the previous neuron and add a constant value. For training this neural network, basically is just finding the right activation function and tuning

the weights between the neurons. However, the neural network is based on numbers instead of text. Therefore, we need to express the text as a number. ChatGPT uses embedding to express all the data, which is a set of numbers.

So basically, ChatGPT will first change the input data into an embedding and then put those embedding into a neural network to get another embedding. Finally, it will generate a new embedding based on the embedding generated by the neural network and transform this new embedding into probabilities for different possible next tokens.

Part III

This article comprehensively explains how ChatGPT works. Basically, the ChatGPT is trying to think and answer like a human, but it does not think like a human. Even though ChatGPT uses a neural network, just like the way humans think, it still generates the answer based on probability. ChatGPT first used a transformer model to change the text into an embedding and then used a neural network to predict what the next words are based on the input data and previous words.